Weather, Class Modality, and Student Willingness During Extreme Conditions

An observational survey of willingness to attend in-person classes under mild, moderate, and severe weather

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Abstract

Students face difficulties attending in-person classes during mild to severe weather conditions such as heavy rain, snow and ice storms, extreme heat or cold, and other hazardous events that affect safety and transportation. This project investigates the belief that severe weather makes students less willing to attend in-person classes and more inclined to switch to online learning by surveying university students about their willingness to attend in person under mild, moderate, and severe weather conditions. Using an observational survey of 13 students, willingness was measured on a 1–5 scale and summarized using descriptive statistics and visualizations. For inferential analysis, ratings were merged into a binary outcome (1-2 = low willingness, 3-5 = high willingness). Then, Cochran's Q test was applied to account for repeated measurements across the 3 types of weather conditions. The test indicated a statistically significant difference in willingness ($p = 0.009404$), with a bootstrap-based power estimate suggesting high power (0.938). These results indicate that students' willingness to attend in-person classes varies significantly with weather severity.

CCS Concepts

• **Applied computing** → **Education**; • **Mathematics of computing** → *Contingency table analysis*; • **Human-centered computing** → Empirical studies in HCI.

Keywords

extreme weather, online learning, student attendance, Cochran's Q test, survey study

1 Introduction

Severe weather such as snowstorms, heavy rain, and heat waves can disrupt campus life and force universities to reconsider how classes are delivered. A common belief is that when weather becomes extreme, students would strongly prefer switching from in-person to online classes to avoid unsafe commutes and discomfort. However, despite the practical importance of this belief for scheduling and safety decisions, there is limited small-scale evidence on how much weather actually changes students' willingness to attend in person.

The central research question of this project is: *How willing are students to switch from in-person to online classes during extreme weather conditions?*Formally, the null hypothesis states that weather conditions have no effect on students' willingness to attend in-person classes, while the alternative hypothesis states that weather conditions do have a significant effect on willingness. Understanding whether this association is present, even in a small convenience sample, can illustrate how quantitative methods help evaluate everyday myths about student behavior.

In this study, weather condition is operationalized as a three-level categorical variable (mild, moderate, severe), and willingness to attend in person is measured on an ordinal 1-5 Likert scale for each weather category. The project uses an observational online survey of university students, descriptive statistics, a contingency table, and the Cochran's Q test to assess whether willingness distributions differ across weather conditions. Although the final sample size is much smaller than originally planned, the analysis illustrates the gap between descriptive trends and formal statistical significance, and provides a basis for discussing implications and limitations.

2 Methods

2.1 Study design and population

The study uses an observational survey design in which students self-report their perceived willingness to attend in-person classes under hypothetical mild, moderate, and severe weather scenarios. There is no experimental manipulation or random assignment; instead, the aim is to describe and analyze naturally occurring variation in responses.

The target population is all university students who attend in-person classes, regardless of institution or location.In practice, the realized sample consists of 13 students who completed the online survey, most of whom are from Stevens Institute of Technology, producing a small convenience sample rather than a representative sample of the population.The initial goal was to collect approximately 30–40 responses to provide adequate variability while remaining feasible for a course project, but time constraints and voluntary participation led to the smaller final sample.

2.2 Participants

Thirteen students participated in the survey, providing complete responses for all questions.The sample included 11 graduate students and 2 PhD students, reflecting a primarily advanced-student population rather than a mix across undergraduate and graduate levels.Most respondents were affiliated with Stevens Institute of Technology and were recruited via social media groups and personal connections, making the sample a convenience sample rather than a representative cross-section of university students.This concentration in a single institution and degree level is important context

when considering how generalizable the results may be. Table 1 describes the demographic characteristics of the survey respondents, including year of study, commute method, class preference, internet access, and home comfort level.

Table 1: Demographic Characteristics of Survey Respondents

Variable	Distribution (N = 13)
Year of Study	11 Graduate Students, 2 PhD Students
Commute Method	Public Transportation (5), Walk (4), Other (4)
Class Preference	Prefer In-Person (6), Prefer Online (5), No Preference (2)
Internet Access	Always Reliable (8), Mostly Reliable (3), Frequently Unreliable (2)
Home Comfort	Very Comfortable (9), Somewhat Comfortable (3), Neutral (1)

2.3 Variables and operationalization

The key independent variable is *weather condition*, categorized into three levels: mild, moderate, and severe.The survey described each category with examples to help participants interpret the scenarios: mild weather included light rain, temperatures around 50–70°F, and light winds; moderate weather included heavy rain, temperatures around 30–40°F or 80–90°F, and moderate winds; severe weather included snow or ice storms, temperatures below 20°F or above 95°F, severe storms, or flooding warnings.

The main dependent variable is *willingness to attend in-person classes*, measured separately for each weather category on a 1–5 Likert scale. A rating of 1 represented "not at all willing / strongly prefer online," 2 indicated slightly unwilling, 3 indicated a neutral stance, 4 indicated willing to attend in person, and 5 represented "very willing / strongly prefer in-person attendance."The survey also collected additional contextual variables to characterize respondents: *Year* (academic level such as Freshman, Sophomore, Junior, Senior, Graduate, or PhD), *Commute* (mode of travel such as walking, driving, public transportation, bicycle/scooter, campus shuttle, or other), *Class_Preference* (overall preference from "Strongly prefer online" to "Strongly prefer in-person"), *Internet_Access* (reliability of home internet), and *Home_Comfort* (comfort and quietness of the home study environment).These variables together capture academic context, commuting situation, and home learning conditions that may influence preferences under different weather conditions.

2.4 Data collection procedure and biases

The data were collected via an online survey created in Google Forms.The survey link was distributed through social media and messaging platforms, including Facebook and WhatsApp groups for Stevens Institute of Technology students, and was later shared with some students from other colleges to broaden the pool.Participation was voluntary and anonymous; respondents could complete the survey at their convenience and were not offered any compensation.

Several sources of bias were identified in the design and collection process. Voluntary response bias is likely, because students with stronger opinions about weather or online learning may have been more motivated to respond.Sampling bias is present since most respondents are from a single institution and from specific online communities, which limits the generalizability of the results to students at other universities or those less active on these platforms.Interpretation bias may occur if participants differ in how they conceptualize "severe weather" despite the examples provided, and nonresponse bias may arise if some students are less comfortable with online forms and therefore underrepresented.These issues are revisited in the limitations section.

Table 2: Contingency table of willingness ratings (1–5) under mild, moderate, and severe weather conditions.

Weather	1	2	3	4	5
Mild	2	3	3	2	3
Moderate	6	1	4	1	1
Severe	9	2	1	1	0

3 Results

3.1 Descriptive statistics

The final dataset contains 13 complete observations and eight variables, including the willingness ratings for each weather category.For willingness to attend in-person classes, the median rating under mild weather was 3, under moderate weather was 2, and under severe weather was 1, indicating that central willingness decreases as weather conditions worsen.The interquartile ranges (IQRs) were 2 for mild, 2 for moderate, and 1 for severe weather, suggesting greater variability in responses under mild and moderate conditions and more consistently low willingness under severe conditions.

Table 2 summarizes the frequencies of each rating (1–5) for each weather category. Under mild weather, ratings are relatively evenly spread across the scale, with counts at all levels from 1 to 5.Under moderate weather, the distribution shifts toward lower ratings, and under severe weather, there is a strong concentration of ratings at 1 and no ratings at 5, visually reinforcing the downward trend in willingness as conditions become more extreme.

A bar chart created in R using `ggplot2` (Figure 1) displays the distribution of ratings for each weather condition side by side. The bars for high willingness (ratings 4 and 5) shrink markedly as conditions move from mild to moderate to severe, while the bar for rating 1 grows, especially under severe weather, highlighting a clear descriptive pattern of reduced willingness to attend in person.

3.2 Cochran's Q test and effect size

We chose the Cochran's Q test because willingness to attend in person was measured repeatedly for the same students under mild, moderate, and severe weather conditions. The observations across weather conditions are therefore not independent, unlike a chi-square test of independence which assumes independent samples. Ratings of 1 and 2 were grouped as low willingness, and ratings of 3, 4, and 5 were grouped as willingness to attend in person. Although rating 3 represents a neutral response, it was grouped with higher ratings to distinguish unwillingness from non-negative willingness. This binarization created a categorical outcome suitable for Cochran's Q test.

The Cochran's Q test yielded a test statistic of $Q(2) = 9.3333$ with a p-value of 0.009404, which is below the significance level= 0.05. This indicates a statistically significant difference in students'

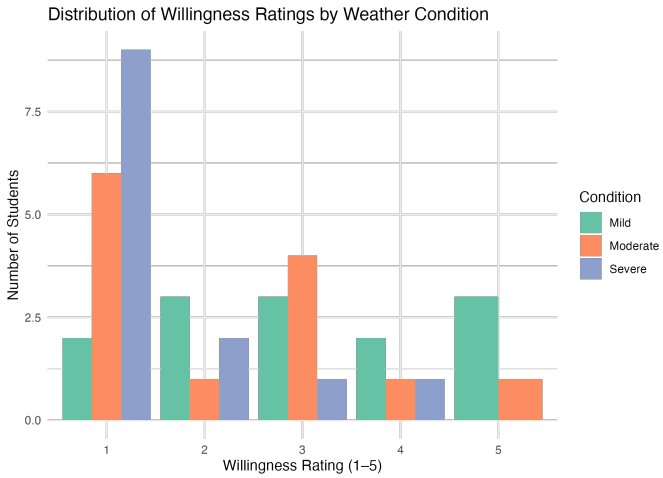


Figure 1: Distribution of willingness ratings to attend in-person classes across mild, moderate, and severe weather conditions. Ratings are shown side by side for each weather category. High willingness ratings (4 and 5) decrease noticeably from mild to severe weather, while low willingness (rating 1) increases, especially under severe conditions, indicating a descriptive pattern of reduced willingness as weather severity increases.

willingness to attend in-person classes across mild, moderate, and severe weather conditions.

A bootstrap-based power analysis yielded an estimated power of 0.938 for the Cochran's Q test. This estimate suggests that, given the observed data, the test had adequate sensitivity to detect differences in willingness across weather conditions, although the result should be interpreted cautiously due to the small sample size.

4 Discussion

4.1 Myth verdict and practical implications

The descriptive statistics and visualizations provide a coherent story: students report lower willingness to attend in-person classes as weather conditions move from mild to moderate to severe, with medians dropping from 3 to 2 to 1 and a clear concentration of rating 1 under severe conditions.This pattern aligns with the intuitive myth that extreme weather reduces students' willingness to attend in person and increases their preference for online options.

Moreover, the results of the Cochran's Q test support the conclusion that students' willingness to attend in-person classes differs across weather conditions. However, because the number of respondents was limited to 13, these findings should be interpreted with caution, and larger and more representative samples are needed to confirm the generalizability of this result.

Despite this, the descriptive patterns suggest that universities should not ignore weather-related preferences, particularly under severe conditions. Even in this small sample, many students reported low willingness to attend in person during severe weather, which supports considering flexible options such as online or hybrid instruction during extreme conditions, while recognizing that larger and more representative studies are needed for firm policy guidance.

4.2 Limitations and future work

Several limitations constrain the conclusions that can be drawn from this analysis. The most important limitation is the small sample size (N = 13). Although the bootstrap-based power estimate was high, the small sample size limits the stability and generalizability of the results.

Measurement and design choices also impose limitations. The willingness ratings were measured on an ordinal 1–5 scale but analyzed with Cochran's Q test that treats categories as nominal, which does not fully exploit the ordering of responses and may reduce the test's ability to detect monotonic trends. The study relies on self-reported perceptions of hypothetical weather scenarios rather than observed attendance behavior during actual weather events, which may differ from real-world decisions. In addition, there were changes in the analysis plan after initial data cleaning, such as reintroducing variables needed for Cochran's Q test, which could have influenced how the data were structured for analysis.

Sampling and bias further limit generalizability. Most respondents are from a single institution, participation was voluntary, and recruitment relied on online student groups, all of which introduce sampling and voluntary response biases. Interpretation bias remains a concern if participants differ in how they understand 'mild', 'moderate', and 'severe' weather, despite provided examples. Future work should aim to collect a larger and more diverse sample across multiple universities and geographic regions, use methods that explicitly respect ordinal response structure (such as ordinal logistic regression), and, where possible, connect survey responses with actual attendance data during documented weather events to more rigorously test the myth.

Acknowledgments

The authors thank the students who participated in the survey and the instructors of CS-810F: Data Analytics and Statistical Methods for guidance and feedback on intermediate project milestones.

References

A Survey Instrument (Excerpt)

The survey included questions about academic level (Year), commute mode, general impact of severe weather on attendance (Weather_Impact_Leve overall class modality preference (Class_Preference), home internet reliability (Internet_Access), and home study environment comfort (Home_Comfort).It also asked students to rate their willingness to attend in-person classes on a 1–5 scale under mild, moderate, and severe weather scenarios described with specific examples.

B R Code (Excerpt)

The analysis was implemented in R using `ggplot2` for visualization and `DescTools::CochranQTest` for inferential testing. The code computed medians and interquartile ranges for willingness ratings under mild, moderate, and severe weather, constructed a 3×5 frequency table of ratings, generated a grouped bar chart

showing rating distributions by weather condition, and then binarized ratings into low (1–2) versus non-low (3–5) willingness. Cochran's Q test was applied to the respondent-level binary data to account for repeated measurements across the three weather conditions. An empirical power estimate for Cochran's Q was computed using bootstrap resampling of respondents by repeatedly re-running the test and calculating the proportion of p-values below $\alpha = 0.05$. The respondent-level binary responses were saved to `respondent_binary_responses_by_condition.csv`.